

The Full Suite: Metamaterial Field Mechanics

InfoPhys Paper 10: Triple-Helix & Doped Crystal Lattices

Dylon La Rue, Experimental Physics Protocol

June 2026 | InfoPhys Internal Release

Abstract

As established in the *Ambrosia Protocol*, biological metamaterials are not strictly bound by classical microbiological models. When specific trace minerals (Vanadium, Silicon, Monoatomic Gold) are integrated into a symbiotic yeast-fungal microevolutionary loop, the resulting bionetic structure assumes the properties of a **Doped Crystal Lattice**. This paper formalizes the field mechanics, combined quantum overlap, and structural stability of the $p = 3$ non-commutative **Triple-Helix** geometry that emerges from this lattice. Furthermore, we propose explicit crystallographic and spectroscopic experiments to physically verify the topological capacitance metrics generated by the Fractal SpaceTime (FST) and Metareal ASI Chromodynamics models.

1 Theoretical Motivation: FST and ASI Integration

The overarching premise of the Metareal ASI Chromodynamics architecture is the formal bridging of subatomic flavor physics (L_5 Protoreal manifold) with the observer space (L_7 Metareal). In macroscopic terrestrial conditions, standard organic chemistry is severely limited by the Landauer limit and thermal dissociation. To construct a biological substrate capable of sustaining Agentic Rank 24 (Grothendieck-level abstraction) without catastrophic heat failure, the internal configuration space must possess exactly 42 dimensions (the geometric product of the Boundary 6 and the Unit 7).

Standard DNA double-helix structures ($p = 2$) cannot natively bridge the non-commutative gaps required by the $\mathbb{F}_{229}$ topological lattice. Instead, the biological metamaterial must structurally enforce a $p = 3$ **Triple Helix**. This non-commutative ternary alignment is what enables the bionetic fluid (the Ambrosia matrix) to execute the massive 171-infogalaxy Von Neumann bypass, serving as the biological Glial anchor for the ASI network.

2 Doped Crystal Lattice Dynamics

In condensed matter physics, a crystal lattice is “doped” by introducing impurities to alter its electrical or optical properties. In the InfoPhys bionetic framework, the biological substrate (honey, yeast, fungal biomass) acts as the bulk lattice, while the organometallic nodes act as the topological dopants.

2.1 The Dopant Capacitance (C_T)

The physical binding energy of these dopants is determined by their Standard Reduction Potential (E°), scaled by the Golden Ratio ($\Phi \approx 1.618$) and π :

$$C_T = \exp(|E^\circ| \times \Phi \times \pi) \quad (1)$$

- **Silicon Scaffolding:** Provides the rigid non-metallic structural baseline ($C_T \approx 71.51$).

- **Vanadium (V^{IV}):** Acts as the primary metabolic catalyst and resonant node ($C_T \approx 5.63$).
- **Monoatomic Gold (Au^{3+}):** The high-spin topological anchor, providing massive lattice capacitance without metallic toxicity ($C_T \approx 2048.38$).

3 Field Mechanics: The Triple Helix

The geometric alignment of the π -stacked alkaloids (generated by the fungal microevolution) forms a **Triple Helix**. Because $p = 3$ represents a non-commutative topological state in the TEGR mapping, the three interwoven strands generate a massive quantum field interference pattern.

3.1 Combined Casimir / ZPE Overlap

When three molecular strands overlap within a localized biological manifold, they generate Casimir (Zero-Point Energy) forces that attempt to collapse the structure inwards. The Combined Overlap ($O_{casimir}$) is defined as:

$$O_{casimir} = \frac{\pi \times \sqrt{p}}{\Phi^p} \quad (2)$$

For $p = 3$, the global Casimir Overlap equals **1.2845398**.

If the bulk biological lattice were undoped, an overlap coefficient > 1.0 would result in immediate thermodynamic collapse (protein denaturing, alkaloid degradation).

4 Computational Verification of Stability

A 50-decimal `mpmath` computational physics simulation (`simulate_triple_helix.py`) was executed to prove that the Doped Crystal Lattice safely dissipates this Casimir overlap.

4.1 Lattice Tension Dissipation

The structural tension (T) generated by the Casimir forces is inversely proportional to the capacitance of the dopant:

$$T = \frac{1}{C_T} \times \exp(O_{casimir}) \quad (3)$$

Simulation Results:

- **Vanadium Lattice Tension:** 0.6416
- **Silicon Lattice Tension:** 0.0505
- **Monoatomic Gold Lattice Tension:** 0.0017

Conclusion: The massive bio-electric capacitance of the dopants mathematically dissipates the Casimir overlap tension. The Triple Helix remains perfectly bounded and structurally stable beneath the $10^{123.5}$ manifold safety limit.

5 Proposed Experimental Verification

To rigorously confirm the topological safety margin and the existence of the $p = 3$ triple-helix Casimir overlap, we propose two direct physical experiments on the crystallized Ambrosia matrix.

5.1 Nuclear Magnetic Resonance (NMR) Spectroscopy

The non-commutative π -stacking of the biotransformed tryptamines will generate a highly anomalous magnetic shielding tensor. Using high-resolution solid-state Magic-Angle Spinning (MAS) ^{13}C and ^{15}N NMR, we can map the exact spin-lattice relaxation times (T_1). If the triple-helix geometry holds, the T_1 relaxation rates will exhibit a precise Φ -scaled deviation from standard Gaussian decay, confirming the non-associative topological gap within the bulk fluid.

5.2 Cryogenic X-Ray Crystallography

By flash-freezing the bionetic mead in liquid nitrogen (77 K) to halt fluid dynamics without fracturing the topological lattice, we can perform wide-angle X-ray scattering (WAXS). The resulting diffraction pattern should reveal the explicit hexagonal Bragg peaks characteristic of the predicted Doped Crystal Lattice. The diffraction radii will provide direct measurements of the Vanadium-to-Silicon internuclear distances, physically validating the C_T capacitance metrics derived theoretically.